CL-II Machine Learning EXPERIMENT NO-01

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

In [3]:
```
import warnings
warnings.filterwarnings("ignore")
```

In [4]:
```
df = pd.read_csv("Wine.csv")

display(df.head())
print("Dataset shape:", df.shape)
print("Columns:", df.columns.tolist())
```

	Wine	Alcohol	Malic.acid	Ash	Acl	Mg	Phenols	Flavanoids	Nonflavanoid.phenols	Pr
0	1	14.23	1.71	2.43	15.6	127	2.80	3.06	0.28	
1	1	13.20	1.78	2.14	11.2	100	2.65	2.76	0.26	
2	1	13.16	2.36	2.67	18.6	101	2.80	3.24	0.30	
3	1	14.37	1.95	2.50	16.8	113	3.85	3.49	0.24	
4	1	13.24	2.59	2.87	21.0	118	2.80	2.69	0.39	

```
Dataset shape: (178, 14)
Columns: ['Wine', 'Alcohol', 'Malic.acid', 'Ash', 'Acl', 'Mg', 'Phenols', 'Flavanoid
s', 'Nonflavanoid.phenols', 'Proanth', 'Color.int', 'Hue', 'OD', 'Proline']
```

In [5]:
```
print(df.info())
print("\nMissing values in each column:")
print(df.isnull().sum())

display(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   Wine                  178 non-null    int64  
 1   Alcohol               178 non-null    float64
 2   Malic.acid            178 non-null    float64
 3   Ash                   178 non-null    float64
 4   Acl                   178 non-null    float64
 5   Mg                    178 non-null    int64  
 6   Phenols               178 non-null    float64
 7   Flavanoids            178 non-null    float64
 8   Nonflavanoid.phenols  178 non-null    float64
 9   Proanth               178 non-null    float64
 10  Color.int             178 non-null    float64
 11  Hue                   178 non-null    float64
 12  OD                    178 non-null    float64
 13  Proline               178 non-null    int64  
dtypes: float64(11), int64(3)
memory usage: 19.6 KB
None

Missing values in each column:
Wine                    0
Alcohol                 0
Malic.acid              0
Ash                     0
Acl                     0
Mg                      0
Phenols                 0
Flavanoids              0
Nonflavanoid.phenols    0
Proanth                 0
Color.int               0
Hue                     0
OD                      0
Proline                 0
dtype: int64
```

	Wine	Alcohol	Malic.acid	Ash	Acl	Mg	Phenols
count	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000
mean	1.938202	13.000618	2.336348	2.366517	19.494944	99.741573	2.295112
std	0.775035	0.811827	1.117146	0.274344	3.339564	14.282484	0.625851
min	1.000000	11.030000	0.740000	1.360000	10.600000	70.000000	0.980000
25%	1.000000	12.362500	1.602500	2.210000	17.200000	88.000000	1.742500
50%	2.000000	13.050000	1.865000	2.360000	19.500000	98.000000	2.355000
75%	3.000000	13.677500	3.082500	2.557500	21.500000	107.000000	2.800000
max	3.000000	14.830000	5.800000	3.230000	30.000000	162.000000	3.880000

In [6]:
```
target_column = "Wine"

# If your CSV uses a different class-column name, change target_column.
X = df.drop(columns=[target_column])
y = df[target_column]

print("Input shape:", X.shape)
print("Target shape:", y.shape)
```

```
Input shape: (178, 13)
Target shape: (178,)
```

In [7]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_scaled_df = pd.DataFrame(X_scaled, columns=X.columns)
display(X_scaled_df.head())
print("Feature means (approximately zero):")
print(X_scaled_df.mean().round(2))

print("\nFeature standard deviations (approximately one):")
print(X_scaled_df.std(ddof=0).round(2))
```

	Alcohol	Malic.acid	Ash	Acl	Mg	Phenols	Flavanoids	Nonflavanoid.p
0	1.518613	-0.562250	0.232053	-1.169593	1.913905	0.808997	1.034819	-0.6
1	0.246290	-0.499413	-0.827996	-2.490847	0.018145	0.568648	0.733629	-0.8
2	0.196879	0.021231	1.109334	-0.268738	0.088358	0.808997	1.215533	-0.4
3	1.691550	-0.346811	0.487926	-0.809251	0.930918	2.491446	1.466525	-0.9
4	0.295700	0.227694	1.840403	0.451946	1.281985	0.808997	0.663351	0.2

```
Feature means (approximately zero):
Alcohol                -0.0
Malic.acid             -0.0
Ash                    -0.0
Acl                    -0.0
Mg                     -0.0
Phenols                 0.0
Flavanoids             -0.0
Nonflavanoid.phenols    0.0
Proanth                -0.0
Color.int               0.0
Hue                     0.0
OD                      0.0
Proline                -0.0
dtype: float64

Feature standard deviations (approximately one):
Alcohol                 1.0
Malic.acid              1.0
Ash                     1.0
Acl                     1.0
Mg                      1.0
Phenols                 1.0
Flavanoids              1.0
Nonflavanoid.phenols    1.0
Proanth                 1.0
Color.int               1.0
Hue                     1.0
OD                      1.0
Proline                 1.0
dtype: float64
```

In [8]:
```
pca = PCA()
X_pca = pca.fit_transform(X_scaled)

print("Original feature shape:", X.shape)
print("PCA-transformed shape:", X_pca.shape)
```

```
Original feature shape: (178, 13)
PCA-transformed shape: (178, 13)
```

In [9]:
```
explained_variance = pca.explained_variance_ratio_
cumulative_variance = np.cumsum(explained_variance)

variance_df = pd.DataFrame({
    "Principal Component": [f"PC{i}" for i in range(1, len(explained_variance) + 1)],
    "Explained Variance (%)": explained_variance * 100,
    "Cumulative Variance (%)": cumulative_variance * 100
})

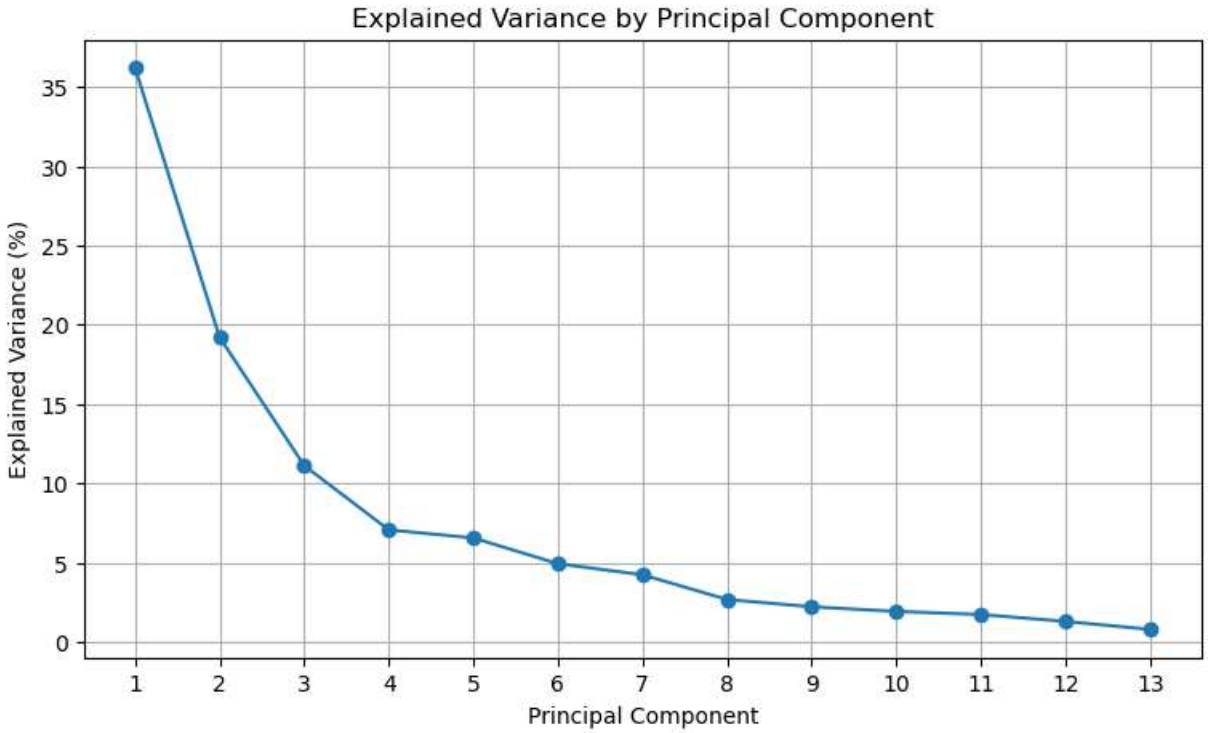
display(variance_df.round(2))
```

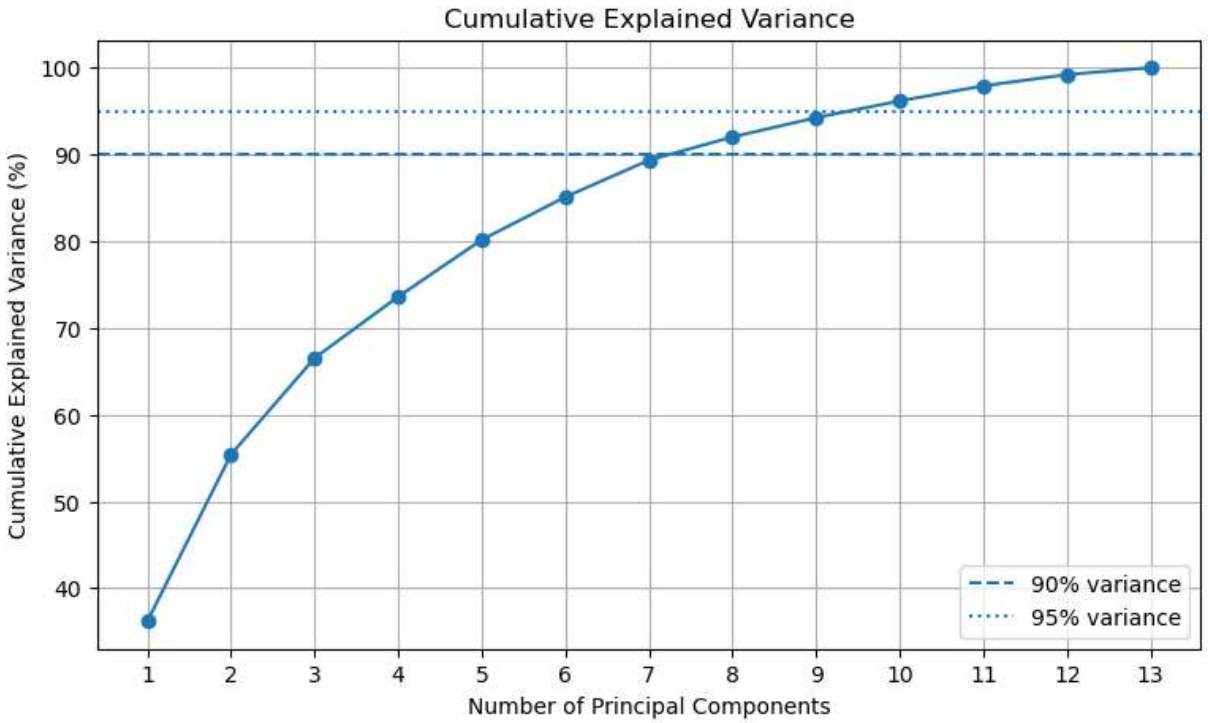
	Principal Component	Explained Variance (%)	Cumulative Variance (%)
0	PC1	36.20	36.20
1	PC2	19.21	55.41
2	PC3	11.12	66.53
3	PC4	7.07	73.60
4	PC5	6.56	80.16
5	PC6	4.94	85.10
6	PC7	4.24	89.34
7	PC8	2.68	92.02
8	PC9	2.22	94.24
9	PC10	1.93	96.17
10	PC11	1.74	97.91
11	PC12	1.30	99.20
12	PC13	0.80	100.00

In [10]:
```
plt.figure(figsize=(9, 5))
plt.plot(range(1, len(explained_variance) + 1),
         explained_variance * 100, marker="o")
plt.xlabel("Principal Component")
plt.ylabel("Explained Variance (%)")
plt.title("Explained Variance by Principal Component")
plt.xticks(range(1, len(explained_variance) + 1))
plt.grid(True)
plt.show()

plt.figure(figsize=(9, 5))
plt.plot(range(1, len(cumulative_variance) + 1),
         cumulative_variance * 100, marker="o")
plt.axhline(y=90, linestyle="--", label="90% variance")
plt.axhline(y=95, linestyle=":", label="95% variance")
plt.xlabel("Number of Principal Components")
plt.ylabel("Cumulative Explained Variance (%)")

plt.title("Cumulative Explained Variance")
plt.xticks(range(1, len(cumulative_variance) + 1))
plt.legend()
plt.grid(True)
plt.show()
```





In [11]:
```
n_90 = np.argmax(cumulative_variance >= 0.90) + 1
n_95 = np.argmax(cumulative_variance >= 0.95) + 1

print("Components needed for at least 90% variance:", n_90)
print("Variance retained:", round(cumulative_variance[n_90 - 1] * 100, 2), "%")

print("\nComponents needed for at least 95% variance:", n_95)
print("Variance retained:", round(cumulative_variance[n_95 - 1] * 100, 2), "%")
```

```
Components needed for at least 90% variance: 8
Variance retained: 92.02 %

Components needed for at least 95% variance: 10
Variance retained: 96.17 %
```

In [12]:
```
pca_2 = PCA(n_components=2)
X_pca_2 = pca_2.fit_transform(X_scaled)

pca_df = pd.DataFrame(X_pca_2, columns=["PC1", "PC2"])
pca_df[target_column] = y.to_numpy()

print("Original number of features:", X.shape[1])
print("Reduced number of features:", X_pca_2.shape[1])
print("Transformed shape:", X_pca_2.shape)

print("PC1 variance:", round(pca_2.explained_variance_ratio_[0] * 100, 2), "%")
print("PC2 variance:", round(pca_2.explained_variance_ratio_[1] * 100, 2), "%")
print("Combined variance:", round(pca_2.explained_variance_ratio_.sum() * 100, 2),

display(pca_df.head())
```

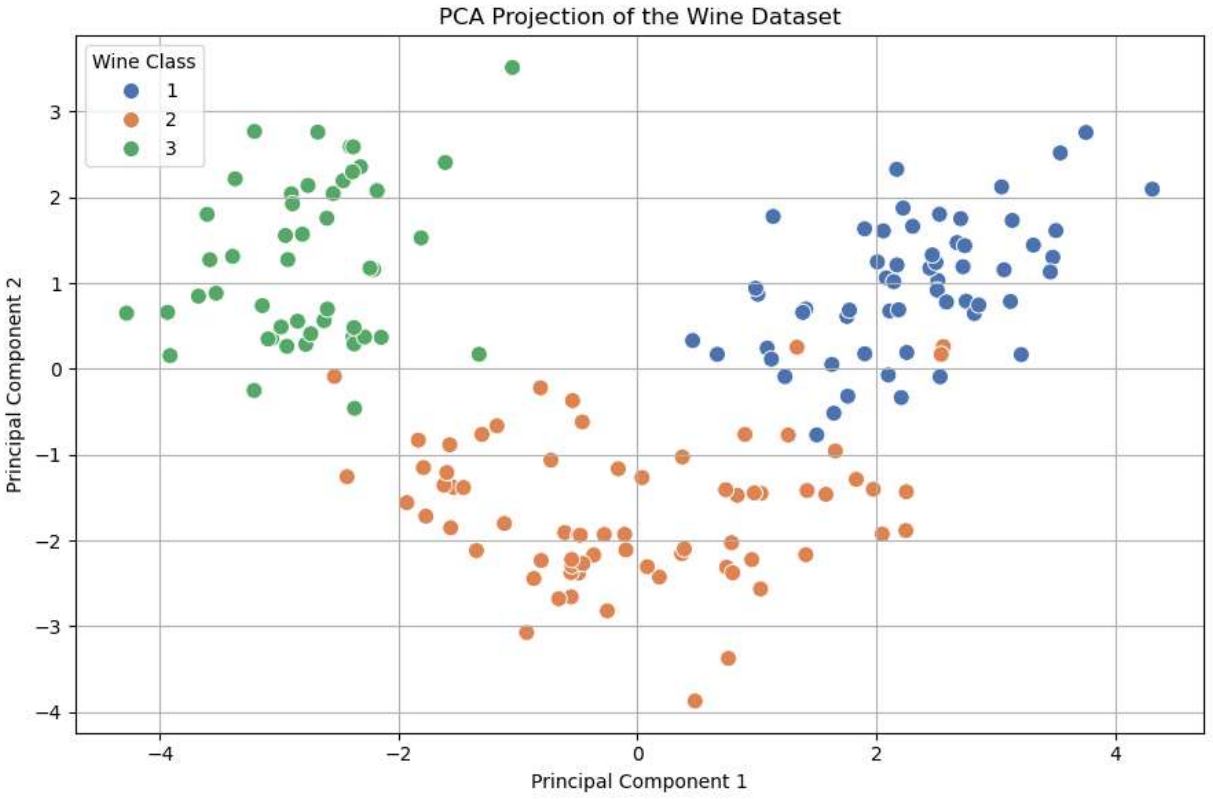
```
Original number of features: 13
Reduced number of features: 2
Transformed shape: (178, 2)
PC1 variance: 36.2 %
PC2 variance: 19.21 %
Combined variance: 55.41 %
```

	PC1	PC2	Wine
0	3.316751	1.443463	1
1	2.209465	-0.333393	1
2	2.516740	1.031151	1
3	3.757066	2.756372	1
4	1.008908	0.869831	1

In [13]:
```
plt.figure(figsize=(9, 6))
sns.scatterplot(
    data=pca_df,
    x="PC1",
    y="PC2",
    hue=target_column,
    palette="deep",
    s=75
)
plt.title("PCA Projection of the Wine Dataset")

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend(title="Wine Class")
plt.grid(True)
```

```
plt.tight_layout()
plt.show()
```



In [14]:
```
loadings = pd.DataFrame(
    pca_2.components_.T,
    index=X.columns,
    columns=["PC1", "PC2"]
)

display(loadings)

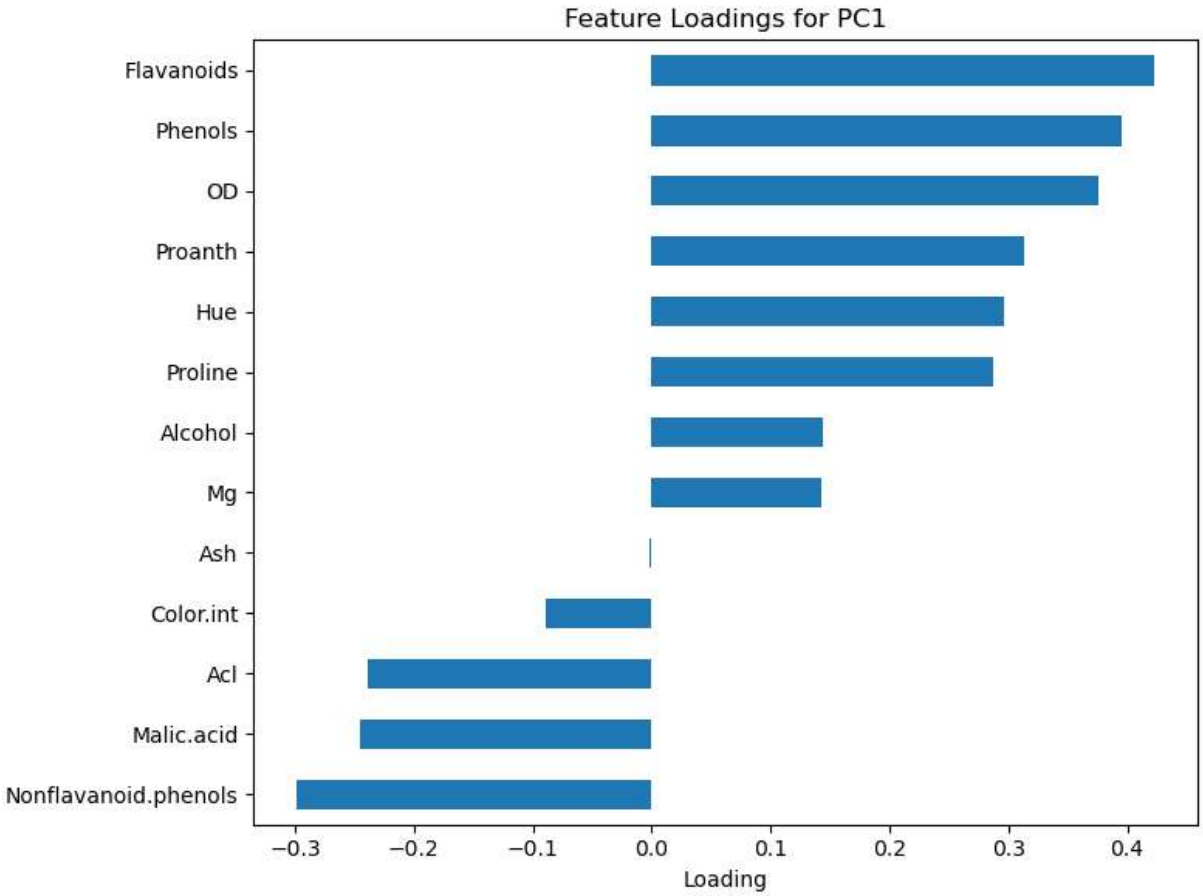
print("Largest absolute loadings for PC1:")
display(loadings["PC1"].abs().sort_values(ascending=False))

print("Largest absolute loadings for PC2:")
display(loadings["PC2"].abs().sort_values(ascending=False))

loadings["PC1"].sort_values().plot(kind="barh", figsize=(8, 6),
                                   title="Feature Loadings for PC1")
plt.xlabel("Loading")
plt.tight_layout()
plt.show()
```

	PC1	PC2
Alcohol	0.144329	0.483652
Malic.acid	-0.245188	0.224931
Ash	-0.002051	0.316069
Acl	-0.239320	-0.010591
Mg	0.141992	0.299634
Phenols	0.394661	0.065040
Flavanoids	0.422934	-0.003360
Nonflavanoid.phenols	-0.298533	0.028779
Proanth	0.313429	0.039302
Color.int	-0.088617	0.529996
Hue	0.296715	-0.279235
OD	0.376167	-0.164496
Proline	0.286752	0.364903

```
Largest absolute loadings for PC1:
Flavanoids              0.422934
Phenols                 0.394661
OD                      0.376167
Proanth                 0.313429
Nonflavanoid.phenols    0.298533
Hue                     0.296715
Proline                 0.286752
Malic.acid              0.245188
Acl                     0.239320
Alcohol                 0.144329
Mg                      0.141992
Color.int               0.088617
Ash                     0.002051
Name: PC1, dtype: float64
Largest absolute loadings for PC2:
Color.int               0.529996
Alcohol                 0.483652
Proline                 0.364903
Ash                     0.316069
Mg                      0.299634
Hue                     0.279235
Malic.acid              0.224931
OD                      0.164496
Phenols                 0.065040
Proanth                 0.039302
Nonflavanoid.phenols    0.028779
Acl                     0.010591
Flavanoids              0.003360
Name: PC2, dtype: float64
```


Feature Loadings for PC1
Flavanoids
Phenols
OD
Proanth
Hue
Proline
Alcohol
Mg
Ash
Color.int
Acl
Malic.acid
Nonflavanoid.phenols
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
Loading